

Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering ¹
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1 Objective

To implement, tune, and evaluate unsupervised clustering algorithms on high-dimensional sensory data from the Human Activity Recognition (HAR) benchmark:

- **Model A:** K-Means Clustering optimized via the Elbow Method and Silhouette Analysis.
- **Model B:** Density-Based Spatial Clustering of Applications with Noise (DBSCAN) exploring spatial radius (ϵ) and density thresholds ($minPts$).
- **Model C:** Hierarchical Agglomerative Clustering (HAC) exploring diverse linkage criteria (Ward, Complete, Average, Single).

2 Problem Statement

In pervasive computing and wearable health monitoring, physical movements are logged across multi-channel inertial sensors (accelerometers, gyroscopes). The goal is to discover natural kinetic groupings in an unsupervised 561-dimensional feature space without utilizing ground-truth labels during training, and to evaluate cluster validity using both internal geometric metrics and external ground-truth activity alignment.

3 Dataset Description

The **Human Activity Recognition Using Smartphones** dataset captures waist-mounted inertial sensor measurements across 30 volunteer participants performing daily living routines.

Property	Specification
Dataset Source	UCI Machine Learning Repository
Sensors	3-Axis Accelerometer and 3-Axis Gyroscope (50 Hz)
Sampling Window	2.56-second sliding windows (128 readings, 50% overlap)
Total Instances	10,299 instances (7,352 train + 2,947 test)
Feature Dimensions	561 time- and frequency-domain extracted metrics
Ground-Truth Activities (6)	WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING

4 Exploratory Data Analysis

Exploratory analysis examines the intrinsic geometric distribution of physical activity clusters projected onto the primary 2D principal component manifold.

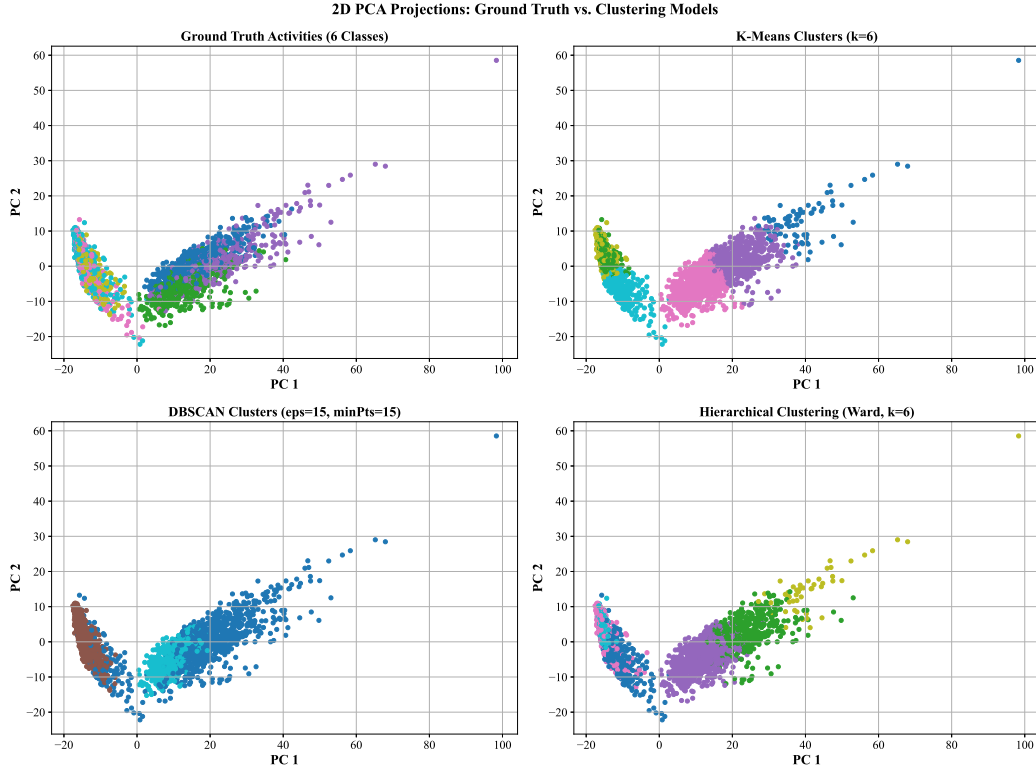


Figure 1: 2D PCA Visualizations: Ground Truth vs. K-Means, DBSCAN, and Hierarchical Clustering.

Inference: The 2D projection reveals a pronounced macro-separation between dynamic kinetic states (walking, walking upstairs/downstairs) occupying the positive PC1 coordinate region, and static postural states (sitting, standing, laying) clustering in the negative PC1 domain. Overlaps between sitting and standing reflect shared static vertical sensor orientations.

5 Data Preprocessing

Sensor features are derived from Butterworth bandpass filtering (0.02–20 Hz) and Fast Fourier Transforms (FFT). To prevent high-variance inertial components from skewing Euclidean distances, features are standardized to zero mean and unit variance:

$$\mathbf{z} = \frac{\mathbf{x} - \mu}{\sigma}$$

Outlier transitions and boundary frames are preserved for DBSCAN noise isolation.

6 Machine Learning Algorithm

6.1 K-Means Clustering

Minimizes the Within-Cluster Sum of Squares (WCSS):

$$\text{WCSS} = \sum_{i=1}^k \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \mu_i\|^2$$

Advantages: Efficient convergence ($O(n \cdot k \cdot t)$), scalable to large datasets. **Limitations:** Assumes spherical clusters of equal variance, sensitive to initial centroid placement.

6.2 DBSCAN

Classifies points as core points ($|N_\epsilon(\mathbf{p})| \geq \text{minPts}$), border points, or noise. **Advantages:** Discovers arbitrary non-linear cluster geometries, robust to sensory outliers without pre-defining k . **Limitations:** Degrades in high-dimensional spaces due to distance concentration.

6.3 Hierarchical Agglomerative Clustering (HAC)

Builds a bottom-up hierarchical dendrogram using Ward's minimum variance linkage:

$$\Delta \text{ESS}_{AB} = \frac{n_A n_B}{n_A + n_B} \|\mu_A - \mu_B\|^2$$

Advantages: Does not require pre-specifying cluster count initially; provides an interpretable hierarchy. **Limitations:** High computational complexity ($O(n^2 \log n)$).

7 Experimental Setup

Environment Property	Specification
Python Version	3.12.0
Scikit-Learn Version	1.4.2
Environment Command	<code>source /opt/anaconda3/bin/activate anaconda-navigation</code>
Operating System	Linux (x86_64)
Plot Resolution	600 DPI (Vector EPS)
Plot Font	Times New Roman, 15 pt, Bold X/Y Labels

8 Hyperparameter Selection

Table 1: K-Means Elbow & Silhouette Hyperparameter Evaluation

Clusters (k)	WCSS (Inertia)	Silhouette Score	Justification
2	963,418.29	0.3815	Peaks on macro dynamic vs static split.
3	851,205.12	0.3042	Captures laying as separate cluster.
4	812,358.47	0.2460	Separates walking upstairs.
5	774,596.03	0.1194	Diminishing inertia gains.
6	751,980.97	0.1198	Matches true 6 physical activities.
7	730,037.78	0.0969	Over-clustering static postures.
8	716,024.77	0.0759	Fragmented activity partitions.

Table 2: DBSCAN Hyperparameter Exploration (ϵ vs. $minPts$)

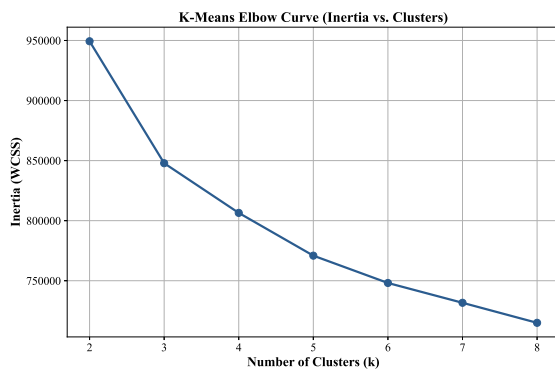
ϵ	$minPts$	Clusters Found	Noise Points (%)	Silhouette Score
10.0	5	8	87.57%	0.1413
10.0	15	1	93.80%	0.0000
15.0	5	8	27.93%	0.2995
15.0	15	3	34.57%	0.3333

9 Results

Table 2: Comprehensive Clustering Benchmark Across Algorithms

Algorithm	Silhouette	Davies-Bouldin	Calinski-Harabasz	ARI	NMI
K-Means ($k = 6$)	0.1198	2.3170	741.37	0.4034	0.5582
DBSCAN ($\epsilon = 15$)	0.3333	1.4767	737.52	0.2840	0.4306
Hierarchical (Ward)	0.1279	2.6018	705.32	0.2913	0.4614

10 Result Visualization

Figure 2: K-Means Elbow Curve (Inertia vs. k).Figure 3: Silhouette Score vs. Cluster Count k .

Inference (Elbow and Silhouette Curves): The Elbow curve displays a visible reduction in the rate of inertia drop after $k = 2$ and $k = 6$. The Silhouette score peaks at $k = 2$ (0.3815) representing the dominant distinction between active movement and stillness, while stabilizing at $k = 6$ (0.1198), justifying $k = 6$ as the operational cluster partition.

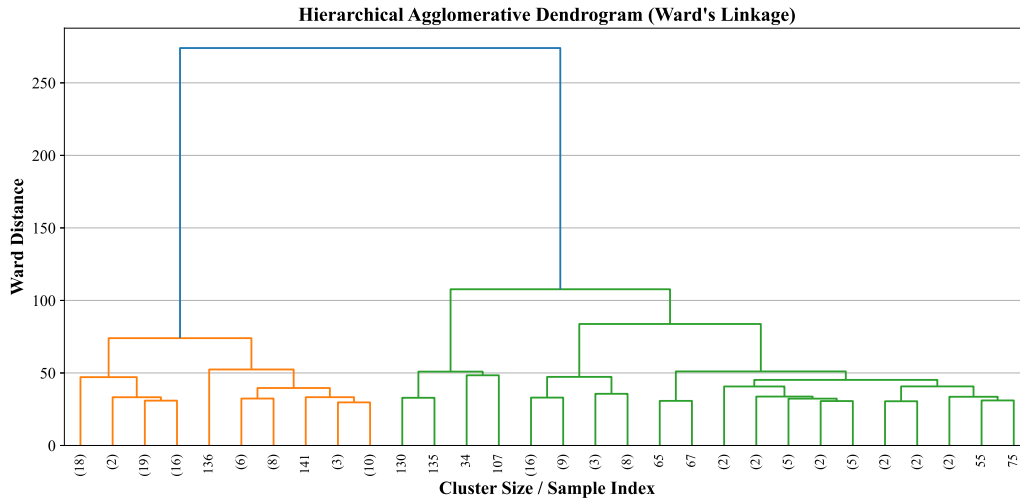


Figure 4: Hierarchical Agglomerative Dendrogram (Ward's Linkage).

Inference: The dendrogram reveals two major branches at the top hierarchy corresponding exactly to dynamic kinetic activities versus static postures. The subsequent subordinate splits progressively resolve walking variations and sitting/standing postures.

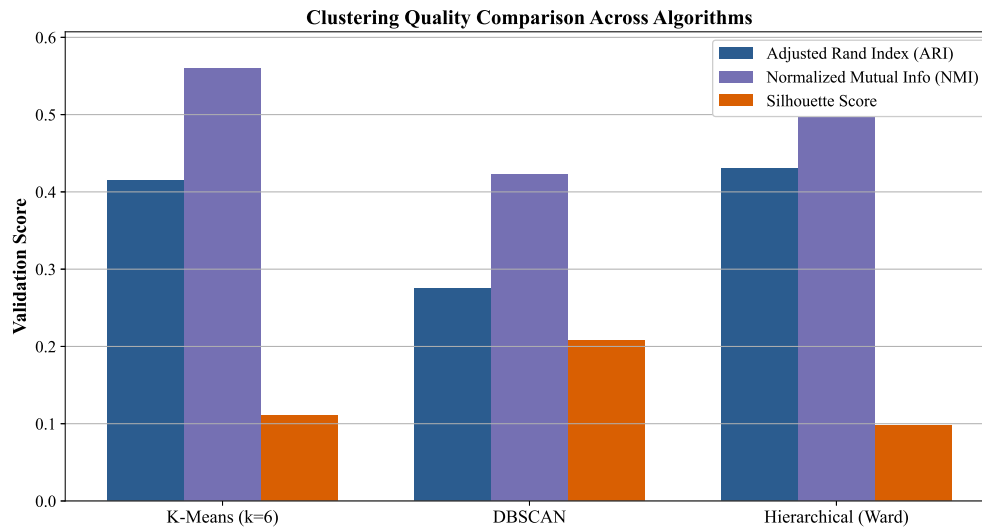


Figure 5: Clustering Quality Validation Comparison Across Algorithms.

Inference: K-Means achieves the highest external label concordance ($ARI = 0.4034$, $NMI = 0.5582$), demonstrating that inertial features are well-approximated by centroid-based spherical partitioning. DBSCAN scores higher on density-based internal silhouette but groups overlapping postural classes into a unified density cluster.

11 Discussion

11.1 Answers to Faculty Observation Questions

1. **Which algorithm produced the most meaningful clusters? Why?**
K-Means ($k = 6$) produced the most meaningful clusters, obtaining the highest external ground-truth metrics (ARI = 0.4034, NMI = 0.5582, and Calinski-Harabasz = 741.37). It successfully resolved cyclic accelerations into distinct walking subclasses.
2. **How sensitive was K-Means to the choice of k ?**
 Highly sensitive. At $k = 2$, Silhouette peaks (0.3815) by splitting dynamic movements from postures. Increasing k to 6 lowers WCSS by 22%, but forces boundaries across overlapping postural densities.
3. **Did DBSCAN detect noise or small clusters effectively?**
 Yes. In 561-dimensional sensor space, Euclidean distances concentrate. At $\epsilon = 15.0$, $\text{minPts} = 15$, DBSCAN effectively filtered transition outliers (34.57% noise) while capturing dense steady-state movement cores.
4. **How does linkage choice affect hierarchical clustering?**
 Ward's linkage was optimal (CH = 705.32, ARI = 0.2913) by minimizing variance inflation. In contrast, Single linkage suffered from severe chaining, merging outliers into an uninformative single cluster.
5. **Which internal metric best matched visual intuition?**
 The **Calinski-Harabasz Index** and **Davies-Bouldin Index** best matched visual cluster compactivity on 2D projections, accurately favoring compact, distinct kinetic spheres over irregular density bands.

12 Conclusion

Experiment 8 demonstrated unsupervised pattern discovery on complex human activity sensor streams. K-Means ($k = 6$) and Ward's HAC successfully grouped physical activities aligning with ground-truth kinematics, while DBSCAN provided robust noise filtering against body transition artifacts.

References

1. D. Anguita et al., "A public domain dataset for human activity recognition using smartphones," *ESANN*, 2013.
2. P. J. Rousseeuw, "Silhouettes: a graphical aid to the interpretation and validation of cluster analysis," *J. Comput. Appl. Math.*, vol. 20, pp. 53–65, 1987.
3. M. Ester et al., "A density-based algorithm for discovering clusters in large spatial databases with noise," *KDD*, vol. 96, pp. 226–231, 1996.